Monte Carlo v. DISORT benchmark for Henyey-Greenstein cloud layer

Validation of https://github.com/sepraca/mc_cloud_rt_visualization Monte Carlo (MC) cloud RT code against DISORT for a homogeneous plane-parallel cloud, no atmospheric gas or aerosol extinction, Henyey-Greenstein (HG) phase function, and a Lambertian reflecting surface.

TOC

I. Numerical Implementations

A. MC implementation

https://github.com/sepraca/mc_cloud_rt_visualization

Settings:

- $N_{photons} = 10^6$
  - 1-σ MC noise ≈ 0.0005 (10^6 photons) absolute for R~0.5, assuming binomial counting statistics [$\sigma^2 = R(1-R)/N_{photons}$]
- To prevent significant cloud side photon escape:
  - Horizontal cloud extent (optical path units) = 150
  - Cloud-base to surface = 0.05 km
- Other optical parameters: as specified in test cases

B. DISORT implementation

PythonicDISORT (Ho, JOSS 2024) v1.7, a Python reimplementation of the Stamnes et al. (1988) discrete-ordinates algorithm, validated by its author against Fortran DISORT to $\sim 10^{-6}$ in fluxes.

https://www.theoj.org/joss-papers/joss.06442/10.21105.joss.06442.pdf

Settings:

- NQuad = 48 streams, 96 HG Legendre coefficients ($\chi_l = g^l$), delta-M scaling, Nakajima-Tanaka corrections for intensities, ω_0 capped at $1-10^{-9}$ for conservative cloud particle scattering cases.
- All fluxes normalized to incident flux $\mu_0 \cdot F_0 = 1$.
- DISORT stream convergence: Reflectance stable to $\sim 10^{-5}$ from 16 to 64 streams; energy closure $R + T_{net} + A_{cld} = 1$ to 2×10^{-8}
- Other optical parameters: as specified in test cases

Verification performed on PythonicDISORT:

1. *Stream (angular) comparisons*: R for Case 1 varies by $<2\times 10^{-5}$ from NQuad = 16 to 64.
2. *Energy conservation*: $R + T_{net} = 1$ exactly (to 2×10^{-8}) for all $\omega_0 = 1$ cases; $F\uparrow = A_{sfc} F\downarrow$ at the base, confirming the implementation of the Lambertian boundary convention.
3. *Vs. independent Monte Carlo* (10^7 photons). Case 1: R = 0.42202 ± 0.00016; Case 2: R = 0.60408 ± 0.00015; Case 7: R = 0.36258 → MC 0.36241 ± 0.00015, T_{net} = 0.25080 vs 0.25087. All within ~1σ.
4. *Intensity-flux closure*: hemispheric integration of the bidirectional intensities recovers the flux values to ~0.5% (expected level of trapezoid quadrature error at 24 polar nodes).

II. Flux and Absorption Comparisons

Fixed input parameters: Cloud Optical Thickness (COT) = 10, HG g = 0.85, Horizontal extent = 150 (optical path), Cloud-base to surface = 0.05 km. The latter two settings are to approximate a plane-parallel cloud layer for comparison with DISORT.

Output:

- Refl. Flux (R) = upward flux (albedo) at cloud-top
- Tran. flux down (T↓) = total downward normalized flux at cloud base
- Tran. flux net (T_{net}) = F↓(τ) – F↑(τ) at base (also equal to normalized flux absorbed by surface)
- Cloud absorptance (A_{cld}) = 1 – R – T_{net}

DISORT v. Monte Carlo (10^6 photons)

Test Case	ω_0	μ_0	Surface albedo	R	T↓	T_{net}	A_{cld}
1: DISORT	1.00	1.0	0.0	0.422	0.578	0.578	0.0
MC				0.423	0.577	0.577	0.0
MC* (1M)				0.423	0.577	0.577	0.0
MC–DISORT				0.001	-0.001	-0.001	0.000
2: DISORT	1.00	0.5	0.0	0.604	0.396	0.396	0.0
MC				0.604	0.396	0.396	0.000
MC–DISORT				0.000	0.000	0.000	0.000
3: DISORT	0.98	1.0	0.0	0.287	0.402	0.402	0.311
MC				0.288	0.401	0.401	0.311
MC–DISORT				0.001	-0.001	-0.001	0.000
4: DISORT	0.98	0.5	0.0	0.449	0.251	0.251	0.300
MC				0.449	0.250	0.250	0.301
MC* (1M)				0.449	0.250	0.250	0.300
MC–DISORT				0.000	-0.001	-0.001	0.001
5: DISORT	1.00	1.0	0.5	0.603	0.794	0.397	0.0
MC				0.603	0.795	0.397	0.000
MC–DISORT				0.000	0.001	0.000	0.000
6: DISORT	1.00	0.5	0.5	0.728	0.544	0.272	0.0
MC				0.728	0.542	0.272	0.000
MC–DISORT				0.000	-0.002	0.000	0.000
7: DISORT	0.98	1.0	0.5	0.363	0.502	0.251	0.387
MC				0.364	0.500	0.251	0.385
MC–DISORT				0.001	-0.002	0.000	-0.002
8: DISORT	0.98	0.5	0.5	0.496	0.313	0.156	0.348
MC				0.496	0.312	0.156	0.348
MC–DISORT				0.000	-0.001	0.000	0.000

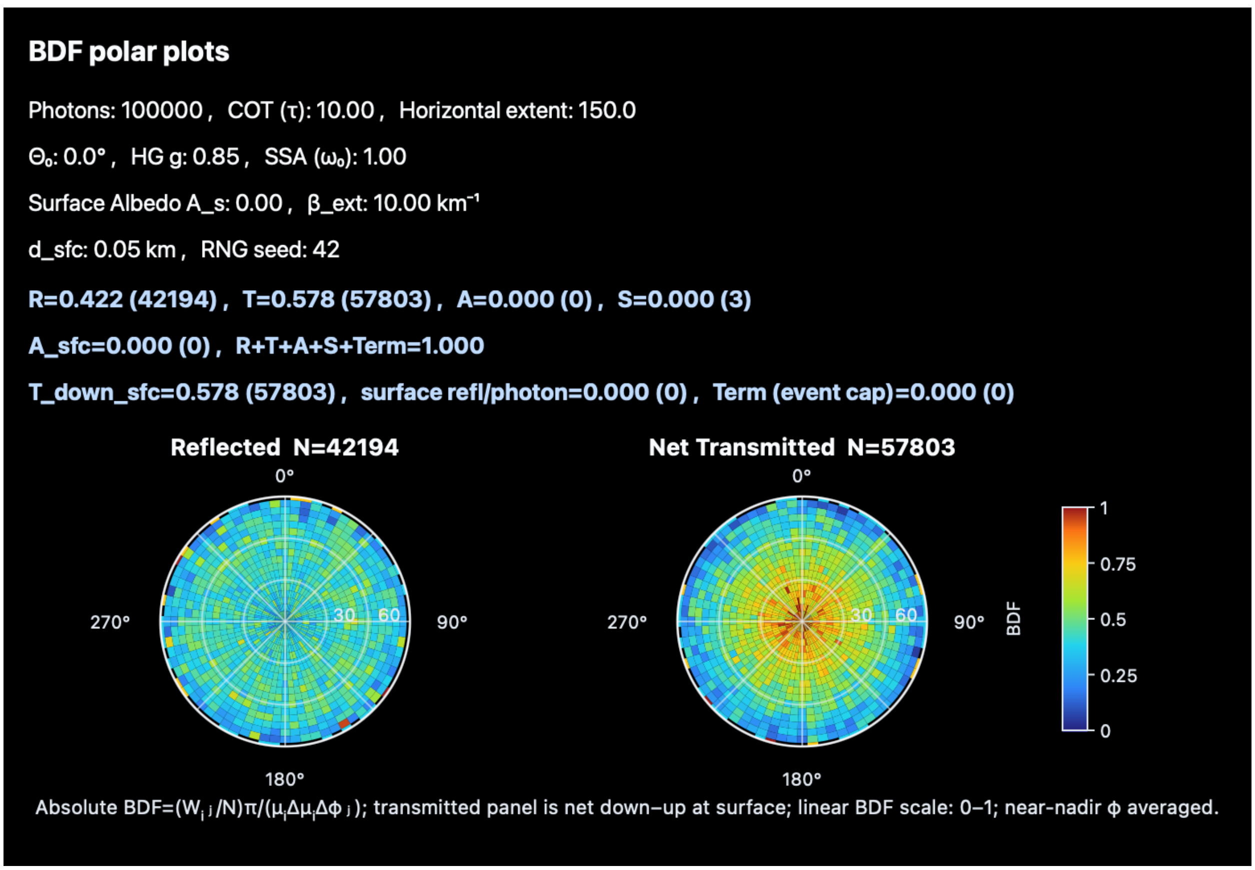
* *MC python code developed and run independent from the html/json code.*

III. Bidirectional Direction Function (BDF) Comparisons

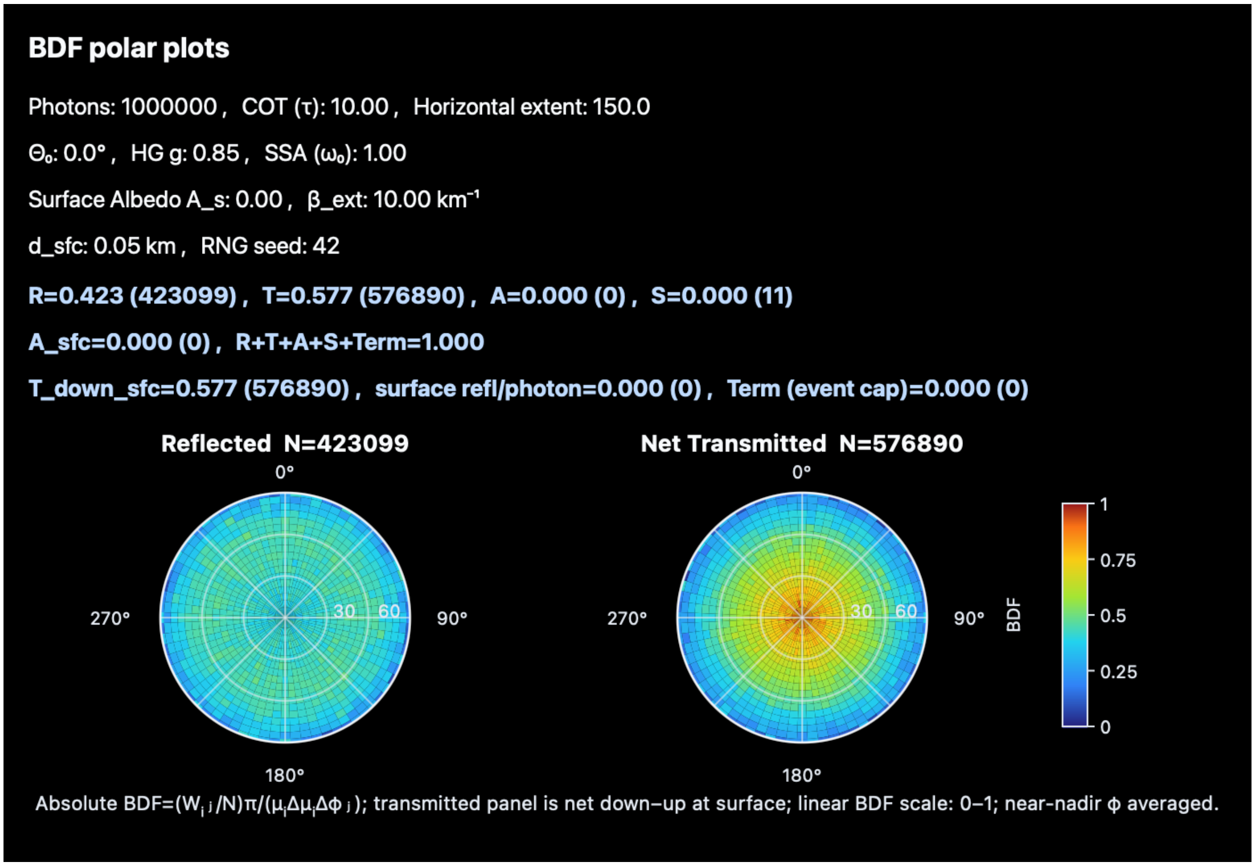
The BDF for a particular i,j bin direction ($BDF_{i,j}$) is given by $\pi I_{i,j}/(\mu_0 F_0)$, consistent with MC code's $(N_{i,j}/N_{photons})\cdot\pi/(\mu_i\ \Delta\mu_i\ \Delta\phi_j)$. DISORT intensities are given at 24 up + 24 down quadrature angles × 72 relative azimuths, NT-corrected. Relative azimuth convention between the two codes is $\phi - \phi_0 = 0°$ is the forward-scattering direction.

Case 1 ($\omega_0 = 1$, $\mu_0 = 1$, $A_{sfc} = 0$) with files *disort_case1_bdf_**

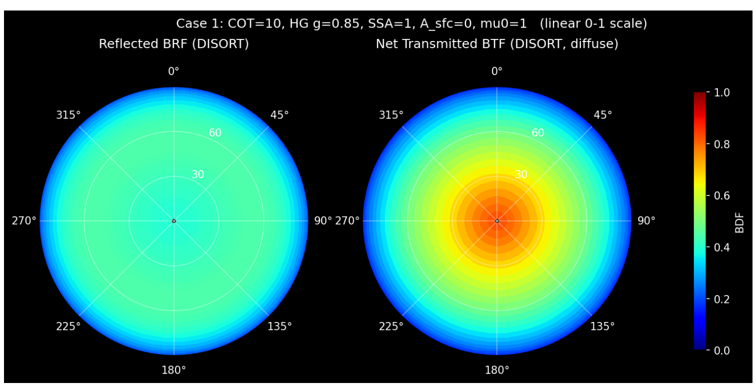
- DISORT field is azimuth-independent machine precision (only the m = 0 Fourier mode survives for $\mu_0 = 1$). Azimuthal speckle in the MC panels is consistent with Poisson noise as expected.
- Reflected BRF: nearly flat, 0.40–0.44, broad maximum near VZA ≈ 55°, falling to ~0.21 at the limb. Matches the MC's uniform green (~0.4–0.5) reflected panel.
- Transmitted BTF: ~0.82 at nadir decreasing monotonically to ~0.17 at the limb. Matches the MC's hotter center that fades to blue edges.



MC with 100K photons, Case 1.



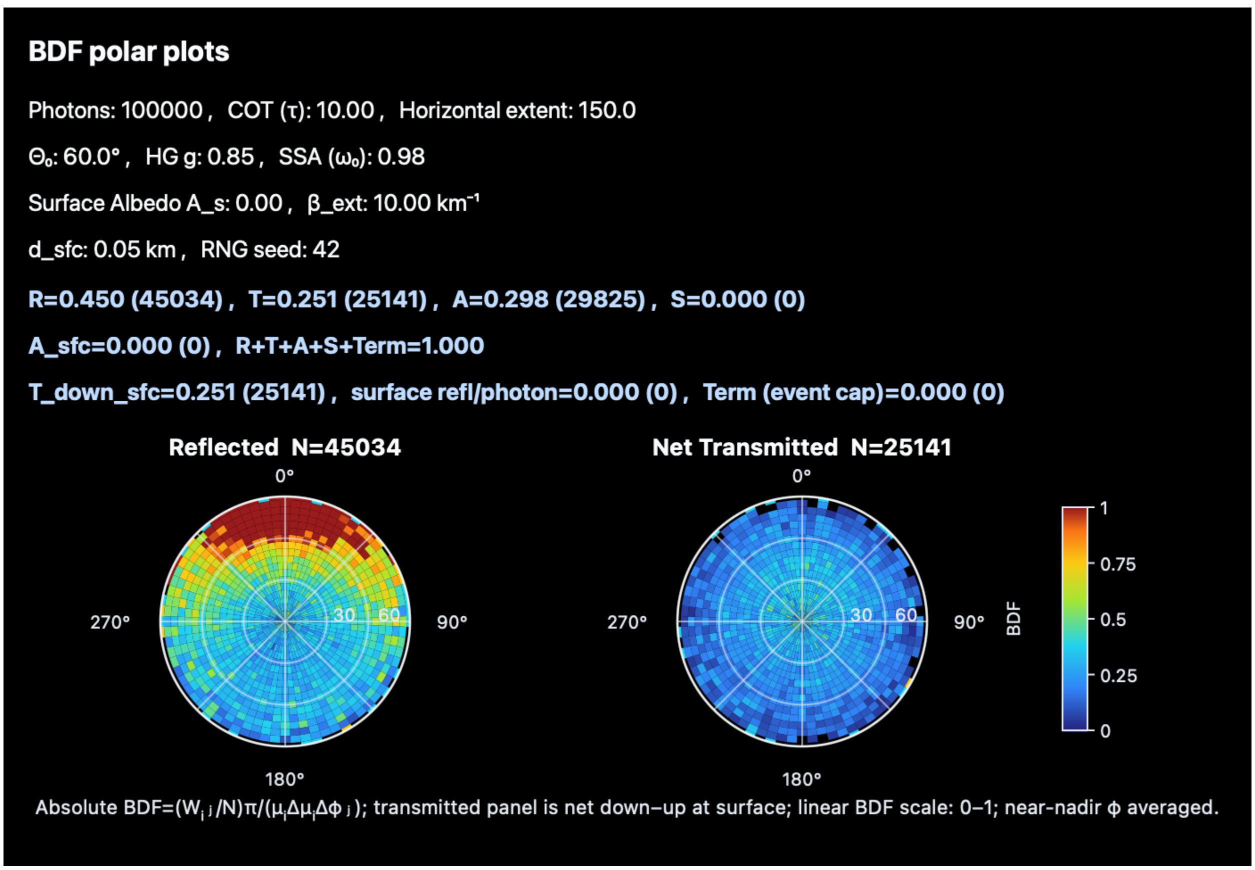
MC with 1M photons, Case 1.



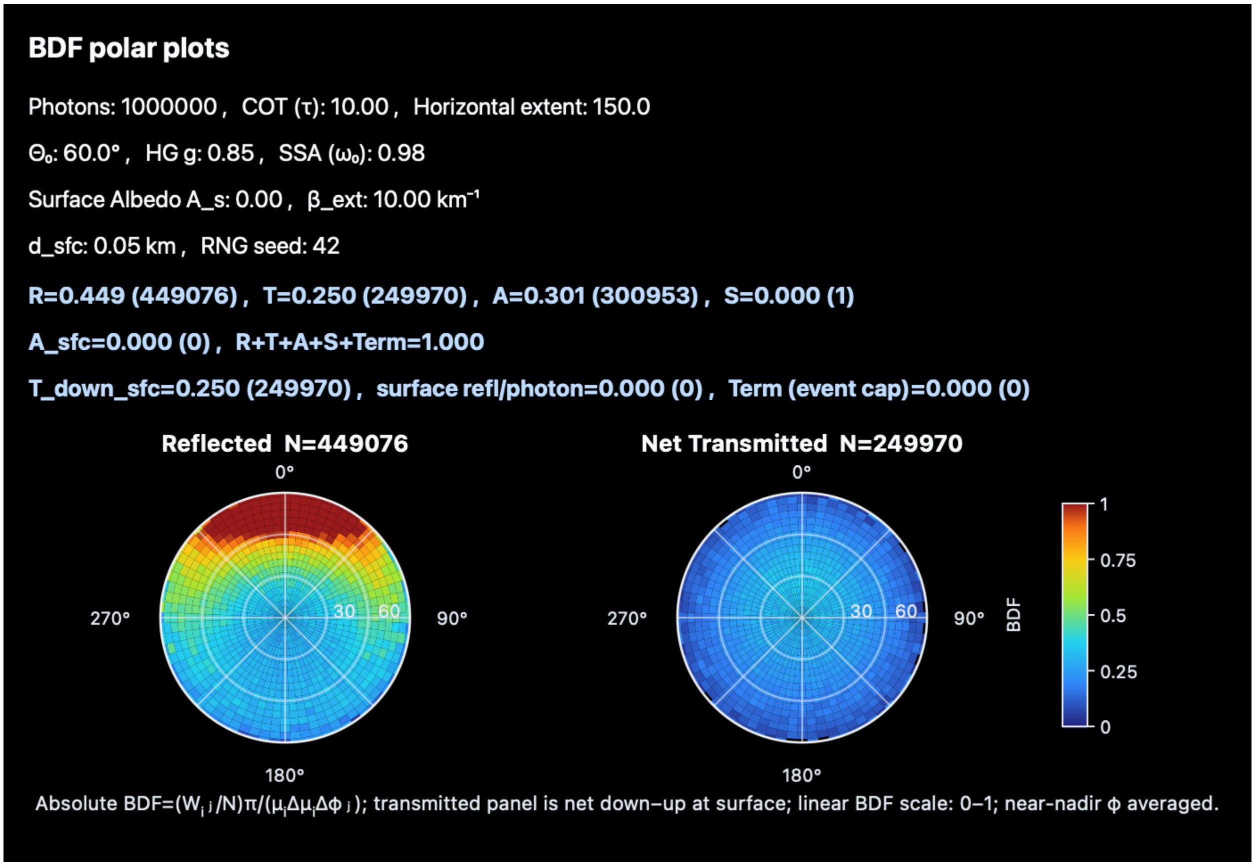
DISORT, Case 1. [Note: not identical color bar v. MC]

Case 4 (ω_0 = 0.98, μ_0 = 0.5, A_{sfc} = 0) with files *disort_case4_bdf_*, disort_case4_brf.csv, disort_case4_btf.csv*

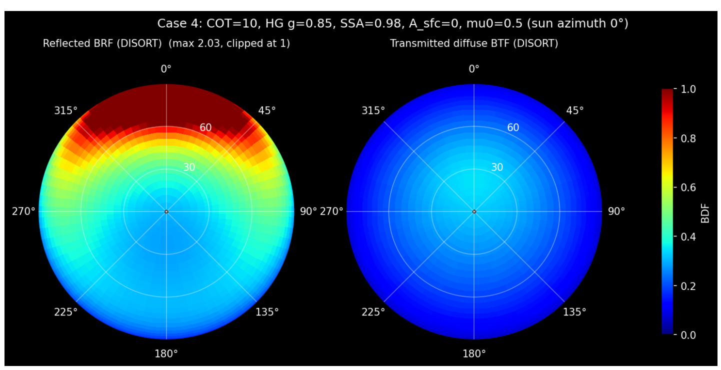
- Integrated quantities from the 10^5 photon MC run (DISORT in parenthesis): R = 0.449 (0.44907), T = 0.252 (0.25053), A = 0.299 (0.30039), giving agreement to ~3 decimal points, which is consistent with expected MC noise.
- Reflected BRF: strong forward-scattering enhancement along ϕ = 0°, rising from ~0.3 at nadir to ~2.03 at VZA ≈ 86°, and exceeding 1.0 beyond VZA ≈ 60° (clipped with the 0–1 color bar used in both figures). The angular extent of the MC's saturated dark-red region matches where DISORT crosses 1.0. Backward azimuth (ϕ = 180°) nearly flat at 0.26–0.31 since HG phase function as no glory/rainbow structure.
- Transmitted BTF: low values (0.06–0.35) and nearly azimuth-independent, slightly brighter at small VZA because multiple scattering for τ=10 with significant absorption reduces solar-azimuth memory.
- Overall, MC results reproduce the BDF structure of the DISORT 48-stream solution, including the forward limb enhancement. Residual differences are visually consistent with Poisson noise (~25K transmitted photons spread over the angular bins).



MC with 100K photons, Case 4.

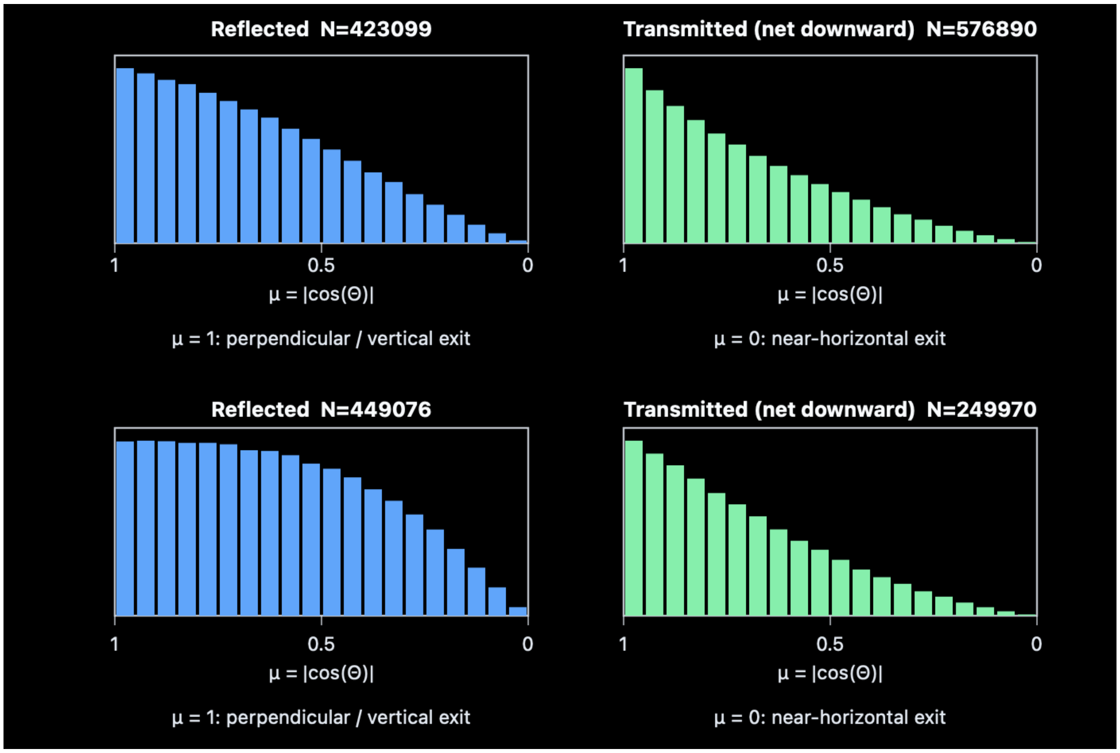


MC with 1M photons, Case 4.

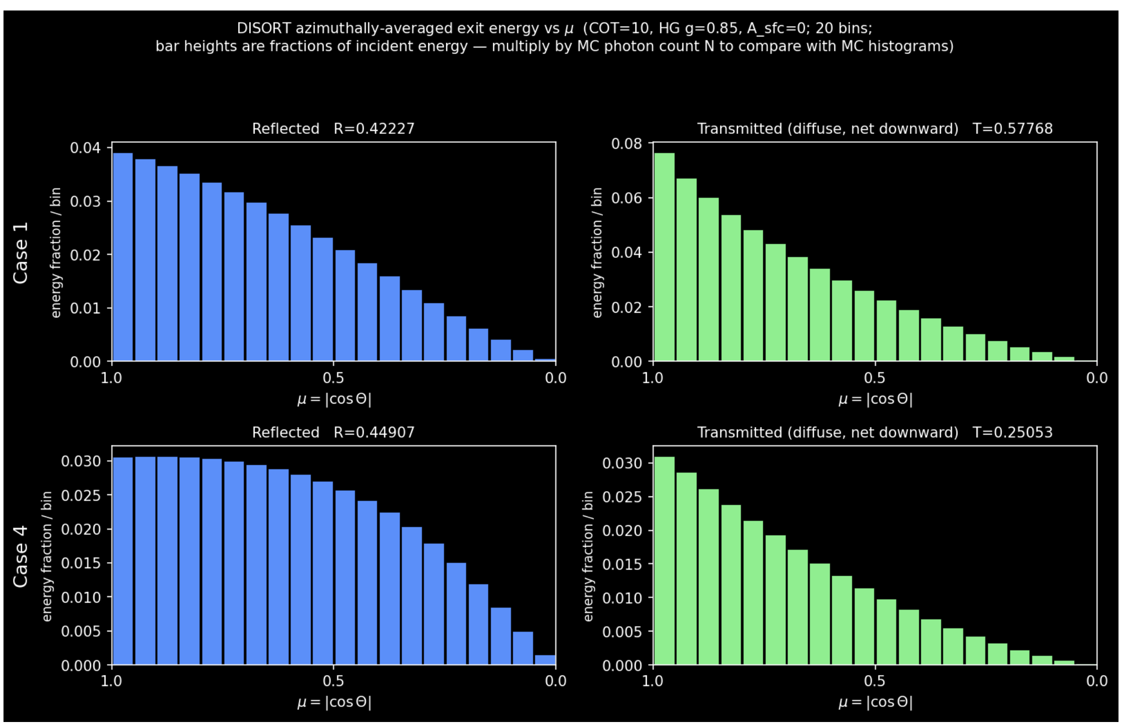


DISORT, Case 4. [Note: not identical color bar v. MC]

IV. View Angle (μ) Histograms

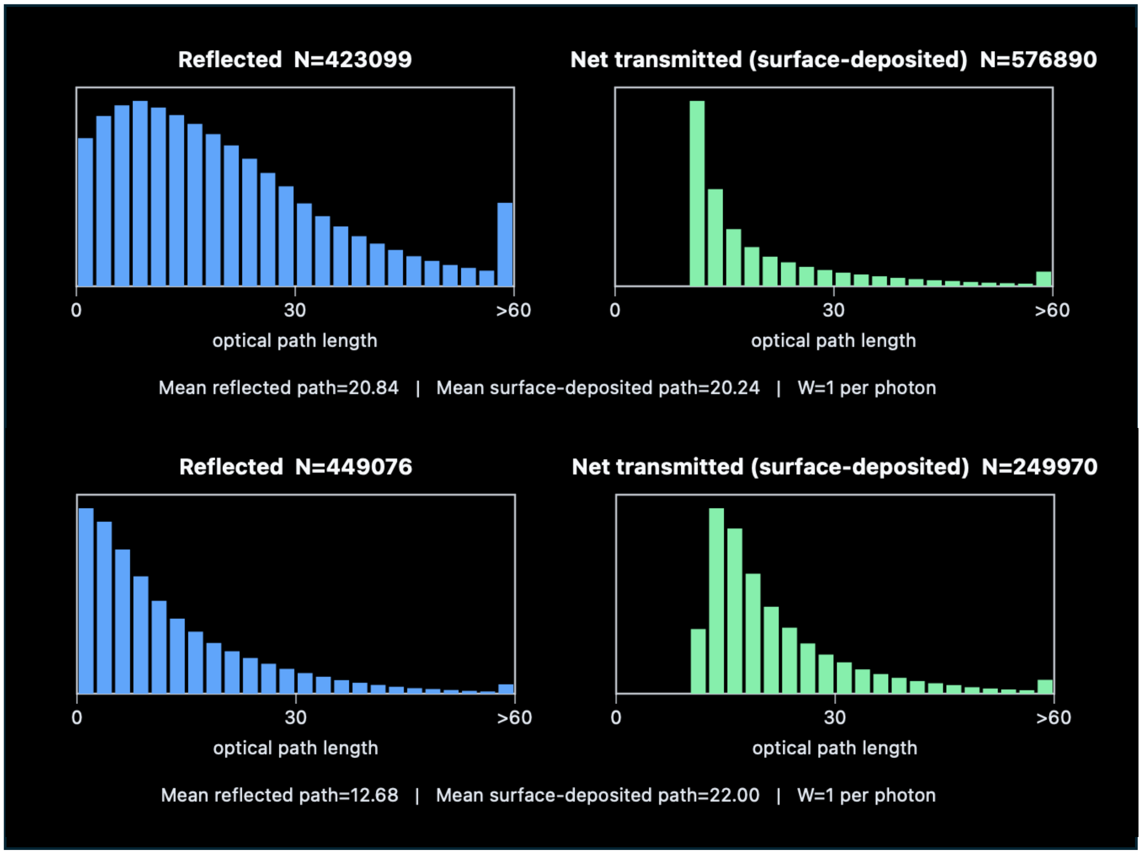


MC view angle histograms with 1M photons. Case 1 (top row) and Case 4 (bottom).



DISORT view angle histograms. Case 1 (top row) and Case 4 (bottom).

V. Other MC Output of Interest



MC path length histograms with 1M photons. Case 1 (top row) and Case 4 (bottom).

VI. Files in this test folder

- *disort_case_table.py*: 8 test cases script
- *disort_hg_test.py*: single-case flux script with stream-convergence loop
- *verify_hg_case.py*, *verify_table_mc.py*: delta-Eddington closed form and independent MC verification
- *disort_case1_bdf.py*, *disort_case4_bdf.py*: bidirectional intensity scripts
- *disort_case1_bdf_polar.png*, *disort_case1_bdf_vza.png*, *disort_case1_bdf.csv*
- *disort_case4_bdf_polar.png*, *disort_case4_bdf_pplane.png*, *disort_case4_brf.csv*, *disort_case4_btf.csv*